Activity Guide — middle / module-09-subtraction-nikhilam

IKS Curriculum

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Activity 01 — Complement & Subtraction Drill

Module: Nikhilam Subtraction · **Band:** Middle · **Day:** 4 · **Time:** 20 min

Type

Paired practice with timed self-correction.

Goal

Each student solves 8 power-of-ten subtraction problems using Nikhilam, with the partner checking accuracy.

Materials per pair

- 1 printed problem sheet (8 problems, see below)
- 1 stopwatch
- 1 pen each
- Pocket calculator (only for the verification round)

Set-up (teacher, before class)

- Print the problem sheet. You can vary numbers — keep difficulty graded from `1000 − N` to `100 000 − N`.

Instructions (student-facing handout)

> **Complement & Subtraction Drill** — work in pairs.

1. Both partners solve all 8 problems individually (8 min). Use the Nikhilam method. Don't look at each other's work.
2. Swap sheets. Partner-A reads problem 1; partner-B re-solves on a fresh sheet (no peeking at A's answer). Compare. If they differ, work out which is correct.
3. Repeat for problems 2–8.
4. Last 4 min: verify any 2 problems using a calculator (number + your complement should equal the base; or compute the subtraction directly).

Problem sheet

```
1.    1000 −  427  = ?
2.    1000 −  892  = ?
3.    1000 −   65  = ?     (pad: 065)
4.   10000 − 3478  = ?
5.   10000 − 1294  = ?
6.   10000 −   87  = ?     (pad: 0087)
7.  100000 − 23 456 = ?
8.  100000 −  7 089 = ?    (pad: 07 089)

Answers (teacher copy):
1.  573        2.  108        3.  935        4.  6522
5.  8706       6.  9913       7.  76 544     8.  92 911
```

Discussion prompts (last 2 min)

- Which problem felt fastest? Why? (Likely #1–3, smaller bases.)
- Which one tripped you up? (Likely #3 or #6 — the padding cases.)
- Did your partner solve any differently than you?

Looks-like / sounds-like

	Looks like	Sounds like
Going well	Pair compares specific digits; debates the padding step	“I think 65 needs to be 065 for this to work”
Needs intervention	One partner just copies; long silences	“Whatever, sure”

Safety / sensitivity

- No public timed competition between pairs. Pairs are self-pacing.

Differentiation

- **Tier up:** Mixed sheet — half power-of-ten subtractions, half “general” subtractions with base-adjustment (preview Day 8).
- **Tier down:** All 8 problems are `1000 − N`. Build fluency at one base before moving up.

Clean-up

- Worksheets collected for teacher spot-check before Day 5.

Activity 1 — Complement Flashcards

Module: Subtraction — Nikhilam (All from 9) · **Band:** Middle · **Duration:** 20 min

Purpose

Reinforce the day-by-day concepts of Subtraction — Nikhilam (All from 9) through a hands-on activity. Aligned with NEP 2020 §4.23 (experiential learning).

Materials

40 flashcards (target − number) pre-made

What students do

In pairs, one student holds up “1000 − 738” (etc.), the other shouts the answer within 3 seconds using the Nikhilam method.

Step-by-step

1. **Setup (5 min)** — gather materials; explain the task.
2. **Working phase (10 min)** — students work individually or in pairs.
3. **Share (5 min)** — selected students share their results with the class.

Differentiation

- **Extension:** ask advanced students to do an additional iteration or analysis.
- **Support:** pair students who need scaffolds; offer partial templates.

Assessment

This activity is **formative**. The teacher observes participation, accuracy, and reflection quality. No grade is assigned.

Safety

- General safety briefing before any materials are handled.
- If herbs, plants, or food are involved: kitchen-amount only, no ingestion of unknown preparations.
- Outdoor activities: stay in the designated area; supervised by the teacher.

Activity 02 — Shopkeeper Role-Play

Module: Nikhilam Subtraction · **Band:** Middle · **Day:** 8 · **Time:** 20 min

Type

Paired role-play with mental computation.

Goal

Each pair runs through 10 “shop” transactions, computing change using Nikhilam. The customer hands a round-denomination note; the shopkeeper returns the change.

Materials per pair

- A small stack of pretend currency: 100s, 500s, 1000s, 2000s (3–4 of each)
- 1 printed scenario sheet (10 items with prices, see below)
- 1 stopwatch
- Whiteboard / scrap paper (allowed but discouraged — push for mental)

Set-up (teacher, before class)

- Print the scenario sheet (vary the items if you like — keep prices in the 50–950 range so all transactions involve a clean base).
- Distribute pretend currency.

Instructions (student-facing handout)

Shopkeeper Role-Play — work in pairs.

1. Decide who's the shopkeeper, who's the customer. Swap after 5 rounds.
2. Customer reads the item + price from the sheet ("One book for 367 rupees").
3. Customer hands over a round note (100, 500, 1000, or 2000) that's larger than the price.
4. **Shopkeeper computes change mentally using Nikhilam.** Says the amount aloud. Hands over corresponding currency.
5. Customer verifies. Mark ✓ or ✗ on the sheet.
6. Time the whole round: 60-second target per transaction.

Scenario sheet

#	Item	Price (₹)	Customer pays (₹)	Change (₹)	Shopkeeper's working
1	Note book	73	100	_____	_____
2	Pen	28	100	_____	_____
3	Lunc h	367	500	_____	_____
4	Book	482	1000	_____	_____
5	Toy	199	500	_____	_____
6	Shoe s	1450	2000	_____	_____
7	Box of sweet s	235	1000	_____	_____
8	Back pack	875	1000	_____	_____
9	T-shirt	549	1000	_____	_____
1	Movi	760	2000	_____	_____

#	Item	Price (₹)	Customer pays (₹)	Change (₹)	Shopkeeper's working
0	e tickets (2)				

```
Answers (teacher copy):
1.  27        2.  72        3.  133       4.  518
5.  301       6.  550       7.  765       8.  125
9.  451       10. 1 240
```

Discussion prompts (last 2 min)

- Which transaction felt the slowest? Why? (Likely #6 or #10 — the 2000-rupee ones; less practised base.)
- Did you give correct change every time? If not, where did the error creep in?
- A real shopkeeper does this all day. How long do you think it took them to get fluent?

Looks-like / sounds-like

	Looks like	Sounds like
Going well	Quick mental call; both partners watching the math	"133. Done."
Needs intervention	Long pause; reaching for paper every time	"Wait… hmm…"

Safety / sensitivity

- Money themes can be sensitive in some classrooms. Pretend currency is fine; avoid framing as "the rich shopkeeper / the poor customer." Keep it neutral.

Differentiation

- **Tier up:** Customer hands a NON-round note (e.g. 870 for a 367-rupee item). Shopkeeper computes `870 − 367 = 503`. This requires the Day-8 general method.
- **Tier down:** Use only 100-rupee notes for all transactions. Build fluency at base 100.

Clean-up

- Currency back in the teacher's box.
- Scenario sheets collected for teacher review.

Extension: project hook

- Pairs whose project is the "shopkeeper's quick-change chart" use today's activity as direct material. Their chart should help a real shopkeeper give change for at least three round-denomination notes.

Project Brief

Project Brief — Module 9 Nikhilam Subtraction (Middle)

Work sessions: Day 8 + Day 9 (in-class time) · **Presentation:** Day 10 · **Total marks:** 40 (project 30 + presentation 10; honesty bonus up to +2)

The project

Each pair (or individual, with teacher approval) chooses ONE of three formats:

Format 1: Shopkeeper’s Quick-Change Chart

A one-page reference card a real shopkeeper could use to give change instantly from 100, 500, 1000, or 2000 rupee notes.

Constraints: - One A4 page (front and back allowed). - Must cover at least 3 round-note denominations. - For each, show the Nikhilam method at least once worked out fully. - Title block with cited source: *Tirthaji, Vedic Mathematics (1965), sūtra Nikhilam.* - Bonus points if the chart is printable AND given to a real shopkeeper for feedback (collect the feedback as a sentence at the bottom).

Format 2: 60-second Teaching Video

A short video that teaches the Nikhilam method to a Grade 4 student.

Constraints: - Total length: 60 seconds (±10 sec). - Must show: the Sanskrit sūtra, at least one worked subtraction (with the answer verified by addition). - Must cite the source (PDF / book + sūtra name) in the title or closing card. - Sound: must be audible.

Format 3: Two’s-Complement Explainer Poster

A poster connecting Nikhilam (base 10) to two’s complement (base 2). Most challenging.

Constraints: - One A2 or larger sheet (or digital equivalent). - Side-by-side comparison: base 10 and base 2. - Worked example in each base. - Title with cited sources: Tirthaji + one CS textbook (Patterson & Hennessy is fine, or your school’s CS book). - A clear final sentence answering: “Why does the same idea work in both bases?”

Why this matters

You’re moving from *learner* to *teacher* (or *engineer* in the poster case). Each role is harder than just doing the problems. Explaining or applying forces you to understand more deeply.

What success looks like

A strong project will: - Demonstrate the Nikhilam method correctly (no arithmetic errors). - Be pitched for the intended audience (a shopkeeper / a 9-year-old / a CS-curious peer). - Show the pair’s own thinking (a unique scenario, a custom example). - Be clearly communicated.

Working in pairs

- Pair work strongly encouraged.
- Individual allowed for the chart or video format.
- Groups of 3 acceptable — each member must have a distinct role.

Citation rules

- At minimum **one** citation must appear in your project — visible.
- Format: *Tirthaji, Vedic Mathematics (1965), sūtra Nikhilam.*
- See `sources.md` for the full list.

Honesty bonus (up to +2)

You can earn up to 2 bonus marks for explicitly addressing the historicity question — i.e., correctly stating that the method is in Tirthaji (1965) and that the deeper “Vedic” provenance is unverified. We reward this because it models the intellectual stance we want.

Self-reflection (submitted with project on Day 10)

In 4–6 sentences: 1. What did you set out to make, and what did you actually end up with? 2. What’s the most surprising thing you discovered while making this? 3. If you had three more days, what would you do differently?

Logistics

- Day 5: project choice submitted.
- Day 8: substantive draft / first artefact.
- Day 9: revisions + presentation rehearsal.
- Day 10: 3-min presentation + submission.